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Module 1

02/01/26 Authentication methods

1. Something you know (knowledge based authentication)
2. Something you have (Token-based authentication)
3. something you are (Biometric based authentication)

Biometric based

1. Physiological biometric trait	2. Behavioural biometric trait
Face, Iris, Retina, fingerprint, palmprint, hand geometry	handwriting, signature, voice, keystroke dynamics, gait (walking patterns)

Characteristics of Good Biometric modality

1. Permanence - modality is stable over a long time
2. Uniqueness - unique among different individuals
3. Universality - every individual should possess
4. collectability - easy to collect
5. Performance - high accuracy and speed
6. Acceptability - whether user is ready to accept to give the verification
7. Resistance to circumvention - resistant or difficult to spoof / forge

Basic component of a Biometric System

1. Sensor - to scan or capture the biometric sample
2. Feature extractor - to extract the distinguishable or essential feature from the sample

3. Template database - where the template is stored
4. Matching Module - Verification process ie matching module will match with the database - whether the person is genuine or imposter.
5. Decision module - check above or below threshold - whether the person is accepted or rejected.

Difference b/w traditional and biometric method

	Traditional Methods	Biometric Methods
Basis	knowledge/possession	Biological trait
Forgettable	Yes	No
Transferable	yes	No
Security	Moderate / low	high
user convenience	Medium	high
risk of loss	high	very low
Implementation cost	low	Moderate to high

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Key Biometric Process

* Enrolment Process
* Recognition Process
  - Verification (1:1 matching)
  - Identification (1:N matching)
  - Template update and Maintenance

Enrolment Process

- first step in any biometric s/m

→ 1. Data capture - user's biometric trait is captured using sensors (camera, scanner, etc.)

→ 2. Quality check - system verifies if the captured data meets the quality standards.

→ 3. Registration - Data is processed and stored in the database.

Eg. enrolling your fingerprint in smartphone - place the finger multiple times on sensor until it captures enough data pts.

Feature Extraction

converting biometric data into a unique digital template

1. Preprocessing - image enhancement, noise reduction, normalization.
2. Feature extraction - identifying unique characteristics (minutiae pts, patterns, ridges)
3. Template creation - converting features into a mathematical representation.

Template storage

Securely storing the biometric template

1. Encryption - Template is encrypted for security.
2. Database storage -
3. Indexing -

4. Matching Process

Comparing live biometric data with stored template.

1. Live capture - new biometric sample is captured
2. Feature extraction - features are extracted from new sample.
3. Comparison - Algm compares with stored template
4. Decision - match or non-match decision based on threshold.

Types of Matching

- Verification (1:1) - confirms claimed identity by comparing one specific template.
- Identification (1:N)
  → 'Are you who you claim to be'
  - user claims an identity & presents biometric slm compares to a single preformed template.

- Identification (1:N)
  → 'who are u'
  - Biometric detail compared with N available samples to find the best match.

Real world application flow

1. Enrolment - traveler iris scanned & registered during 1'st visit.
2. Feature extraction - slm extracts unique iris pattern & create template.
3. Storage
4. Matching - on ~~normal~~ next visit, iris is scanned & matched for instant verification.

Key challenges.

1. Environmental factors
lighting quality, noise, weather affecting capturing quality
2. User Cooperation
Proper position, movement, user behaviour affecting accuracy.
3. Security threats
Spoofing attacks, fake biometrics, replay attacks
4. Privacy attacks
Data protection unauthorized access, correct management.

Performance Measures

- How well a biometric system works.
- System quality: Accuracy & reliability of the sm
- User experience: Balance security with convenience
- security level: determine sm security strength.
- Sm comparison: compare diff biometric technologies.

Basic Matching outcomes

- True accept - genuine user correctly accepted
- False accept - Imposter wrongly accepted.
- False reject - Genuine user wrongly rejected
- true reject - imposter correctly rejected.

False accept rate (FAR)

- % of times an imposter is incorrectly accepted as genuine user.

- It is also called false match rate (FMR).

  FAR = (false accepts / total imposter attempts) × 100

eg: fingerprint sln tested with 1000 imposter attempts.
5 imposter were incorrectly accepted
FAR = (5/1000) × 100 = 0.5 %.
Lower FAR = Better Security (fewer imposter get through)

False Reject Rate (FRR)

- % of times a genuine user is incorrectly rejected.
- It is also called False non match rate (FNMR)

  FRR = (false reject / total genuine attempts) × 100.

- lower FRR = Better user experience

trade-off b/w FAR & FRR

- FAR & FRR work against each other.
- when you decrease one, other increase.

high security setting	high convenience setting
• very strict match	• Lenient matching
• low FAR (secure)	• low FRR (convenience)
• high FRR (inconvenience)	• high FAR (less secure)
• eg: Bank vault access	• eg. phone unlock

Equal Error Rate (EER)

- The point where FAR equals FRR.

why it is important:

- single number to compare diff s/m.
- shows optimal balance point.
- Lower EER = better overall s/m.

eg. comparison:

s/m A: Iris scanner EER = 0.5%

" B: fingerprint EER = 2%

" C: face recognition EER = 5%

→ s/m A (Iris) performs best overall.

Genuine Accept Rate (GAR)

- % of time genuine users are correctly accepted.
- also called True Accept Rate (TAR)

→ GAR = 100 − FRR

→ higher GAR = Better user satisfaction.

eg: Iris recognition s/m at office

1000 genuine employee try to enter

980 are correct accept on 1'st attempt

FRR = (980/1000) × 100 = 98%

GAR = 100 − 98 = 2%

Failure to Enroll Rate (FTE)

- % of users who cannot be enrolled in s/m

FTE = (failed enrollments / total attempts) × 100

→ Common reasons for failure:
- Poor quality biometric traits
- Phy disabilities or injuries
- Environmental factors during capture
- non-cooperation or improper positioning

→ eg: ~~Low~~ fingerprint s/m in construction company
100 worker attempt enrolment
3 worker cannot enrol due to worn fingerprint
FTE = (3/100) × 100 = 3 %
* Lower FTE = more inclusive s/m

Performance matric summary

	what it measures	Lower is better ?
FAR	imposter wrongly accepted	yes
FRR	genuine user wrongly rejected	yes
EER	pts where FAR = FRR	yes
GAR	genuine user correctly accepted	no (higher better)
FTE	user who cannot enroll	yes

* A good biometric s/m balances all these metrics to provide both high security and good user experience.

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Assessing privacy risks of biometrics

- As the biometric traits are permanent & can't be changed, it should be protected.
- misuse can lead to identity theft and surveillance.

* Types < physiological, behavioural } Each type has diff privacy application

Privacy vs security

- security protects s/m from attacks.
- privacy protect use identity & personal data
- A secure s/m can still violate privacy.

Key Privacy risks in Biometrics

- Irrevocability of biometric traits.
- Data breaches & unauthorized access
- Function creep & cross matching
- Mass surveillance.

Risk: Irrevocability

- Biometric traits can't be replaced like passwords
- Compromise causes permanent damage
- Eg: Leaked Fingerprint data

Biometric data breaches

- centralized databases attract attackers.

- stolen templates can be misused.
- Large scale breaches affect millions.
- Every biometric modality has a unique feature. In fingerprint, the features are patterns. These features are stored as templates in DB.

Function creep.

- Data reused beyond original purpose
- often w/o user content
- Violates data protection principles
- Eg: face recognition details are reused somewhere else

Cross matching

- Linking biometric data across s/m
- Enables tracking & profiling
- Reduces anonymity

Surveillance

- facial recognition enables mass surveillance
- used in public spaces w/o consent
- threatens civil liberties.

Assessing privacy risks

- Identify biometric modality used.
- Analyze data collection & storage.

- Evaluate sharing and access policies
- Assess impact on user rights.
- Every biometric sys has their own standards and certain policies are followed by them.

Technical factors affecting privacy
centralized vs decentralized storage
template protection technique
Encryption and access control

Legal and ethical considerations
user content and transparency
purpose limitation
Compliance with data protection law.

Privacy protection techniques
*) Cancelable biometric, *) Biometric crypto system, *) Encrypted & on-device storage.

Case study: National Biometric ID sys
- large scale biometric data collection. High privacy & security risks.
- requires strong safeguards.

Best Practices
- collect minimal biometric data
- use privacy-by-design approach.
- ensure legal compliance & audits.